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SAFETY DISCONNECT CIRCUIT WITH AUTOMATIC RETRY

Abstract

Described embodiments include a safety disconnect circuit having a logic circuit with first, second and third logic inputs and a logic output. A latch has a latch input coupled to the logic output, a latch power input, and a latch output. A counter has a counter input coupled to the logic output, and a counter output. A switch is coupled between a fault terminal and a ground terminal, and has a switch control terminal coupled to the latch output. A first voltage supply has a first supply input and a first supply output. A second voltage supply has a second supply input and a second supply output. A multiplexer has first and second multiplexer inputs that are coupled to first and second supply outputs, respectively. The multiplexer output is coupled to the latch power input. The multiplexer control terminal is coupled to the counter output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/559,510 filed Feb. 29, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] This description relates to safety disconnect circuits. Safety disconnect circuits are sometimes included in DC-DC voltage converter circuits. Safety-critical systems may require provisions for a safety disconnect in the event of a catastrophic fault within the system. A safety-critical system is a system that needs to function properly in order to avoid a worst-case scenario, such as human injury and/or death, significant property damage, financial loss, environmental damage, or a catastrophic systemic effect. Examples of a safety-critical system include automobiles having autonomous driving control, motor control systems in factories, and anywhere a person could potentially get seriously injured by high voltage.

[0003] One example of a system implementing a safety disconnect circuit is a buck DC-DC voltage converter. A potential failure mode in a buck DC-DC voltage converter is a short circuit in the high side drive FET, which can potentially apply a high voltage directly on the output voltage terminal of the DC-DC voltage converter. If this occurs, it can cause damage to downstream equipment connected to the voltage regulator, which can lead to further failures.

[0004] A safety disconnect circuit usually includes a safety mechanism that detects the failure, then takes the appropriate actions to protect the system and its users. In the case of a DC-DC voltage converter, the appropriate action may be to disconnect and latch off power to the system until the failure can be remedied. If the safety disconnect circuit is in an automobile, the owner of the automobile may be required to take it to a dealer for repairs before the power can be unlatched and the system returns to normal operation. To avoid an unnecessary and potentially large inconvenience to the owner, it is important that the safety disconnect circuit avoids false reports of a fault in order to avoid an unnecessary power disconnect and latch-off.

SUMMARY

[0005] In a first example, a safety disconnect circuit includes a logic circuit having first, second and third logic inputs and a logic output. A latch has a latch input, a latch power input, and a latch output. The latch input is coupled to the logic output. A counter has a counter input and a counter output, wherein the counter input is coupled to the logic output. A switch is coupled between a fault terminal and a ground terminal, and has a switch control terminal. the switch control terminal is coupled to the latch output.

[0006] A first voltage supply has a first supply input and a first supply output. A second voltage supply has a second supply input and a second supply output. A multiplexer has first and second multiplexer inputs, a multiplexer control terminal, and a multiplexer output. The first multiplexer input is coupled to the first supply output. The second multiplexer input is coupled to the second supply output. The multiplexer output is coupled to the latch power input, and the multiplexer control terminal is coupled to the counter output.

[0007] In a second example, a safety disconnect circuit includes a latch having a latch input, a latch power input, and a latch output. The latch input is coupled to a validation terminal. A counter has a counter input, a counter supply terminal and a counter output. The counter input is coupled to the validation terminal. A first switch is coupled between a fault terminal and a ground terminal, and has a first switch control terminal. The first switch control terminal is coupled to the latch output.

[0008] A second switch is coupled between a first voltage supply terminal and the latch power input, and has a second switch control terminal. The second switch control terminal is coupled to the counter output. A third switch is coupled between a second voltage supply terminal and the latch power input, and has a third switch control terminal. The third switch control terminal is coupled to the counter output.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows a block diagram for an example safety disconnect circuit with automatic retry following an initial failure.

[0010] FIG. 2 shows a schematic diagram for an example circuit for toggling the enable terminal of a safety disconnect circuit in response to the nFAULT signal.

[0011] FIG. 3 shows a schematic diagram for an example safety disconnect circuit wherein the enable terminal is toggled in response to the nFAULT signal.

DETAILED DESCRIPTION

[0012] In this description, the same reference numbers depict same or similar (by function and/or structure) features. The drawings are not necessarily drawn to scale.

[0013] In DC-DC converters, a circuit failure can lead to a condition where the output voltage or the output current is outside of an acceptable range, or a temperature rise that can affect the safety of the end user or the equipment. Numerous standards have been developed to address the required safety levels for different types of end equipment. In many cases, these safety standards require the system to detect and respond appropriately to failures within a particular amount of time.

[0014] For example, in the case of a DC-DC buck voltage converter, the failure of a pass switch (e.g. high-side switch) or a voltage feedback terminal that is shorted to ground can cause the entire input voltage to be passed to the output voltage terminal of the device. In many cases, this may cause damage to the downstream electronics that can lead to catastrophic failures in the system. In order to prevent such catastrophic failures, a separate battery voltage compliant external supervisor may be used to detect an overvoltage at the output voltage terminal, and respond by disconnecting the input voltage source that is providing power to the DC-DC buck voltage converter and/or the output voltage terminal that is providing power to downstream electronics.

[0015] However, this circuitry can add significant cost due to the additional components that may be required to implement the power-down circuitry. Furthermore, the additional circuitry can significantly increase the quiescent current drawn in the system, which is an undesirable outcome. In an automotive application, increasing the system quiescent current can drain the battery more quickly while the automobile is idle.

[0016] After the fault latch is set, the power-down condition remains in effect until the fault latch is reset. In some cases, resetting the fault latch may require returning the equipment to the manufacturer or a repair shop. It is possible for the power-down latch to be triggered by system noise and/or other transient conditions that may resolve themselves quickly and without intervention. However, a fully latched circuit would not be able to recover from the temporary and non-persistent or intermittent transient condition if the circuit has been powered down. This results in a false detection of a nonexistent fault leading to an unnecessary system power shutdown.

[0017] An important consideration is determining the proper criteria for the safety mechanism to trigger a system shut down. If the safety system triggers falsely, it can cause the system to react to a failure that does not actually exist, possibly causing a properly working system to be shut down unnecessarily. So, it may be advantageous to recheck the system following an initial failure report to verify that the failure is real, and that the failure is persistent and will not resolve on its own.

[0018] It is advantageous to have a safety disconnect circuit with a retry feature that allows the

system to automatically retry following an initial failure. Power is removed from the system during the retry to prevent damage in case the fault is real. Retrying before declaring a permanent failure may help prevent the system from being put into a latched off state in response to a false trigger. Then, if the unsafe condition rectifies itself and the fault does not occur again during the retry, the system will continue with its normal operation.

[0019] In the case of an automotive application, this may prevent unnecessarily returning a car to a dealership or repair shop to clear the fault latch and turn the power back on in a situation where the automobile would have been safe to operate without any repairs. Automatically retrying up to a maximum number of retries allows the system to recover from an intermittent or non-persistent failure, and gives the system the opportunity to check multiple times to verify a fault before locking the system. This helps to ensure the safety system is not locking out power in response to an intermittent fault condition that clears itself while continuing to protect the system during the retries.

[0020] FIG. 1 shows a block diagram for an example safety disconnect circuit with automatic retry **100**. Fault sensors (not shown) monitor particular system parameters such as voltages, currents, and temperatures. Each fault sensor provides a respective fault signal. A first fault signal, fault1 **102**, is provided to a first input of logic circuit **108**. A second fault signal, fault2 **104**, is provided to a second input of logic circuit **108**. A third fault signal, fault3 **106**, is provided to a third input of logic circuit **108**. Logic circuit **108** may have more or less than three fault signals provided as respective inputs.

[0021] Logic circuit **108** provides a signal assertFault **110** at its output that is asserted if any of the fault signals fault1 **102**, fault2 **104**, or fault3 **106** indicate a fault condition. Logic circuit **108** includes logic and timing circuits. Each of fault signals fault1 **102**, fault2 **104**, and fault3 **106** are provided to an individual timing circuit specific to the type of fault sensor that is providing each respective input. For certain types of faults, the system may need to respond and validate the signal immediately with no delay. However, with other types of faults, a timing circuit may ensure that the fault remains for a period of time before validating the fault signal as indicating a legitimate fault. Each of the validated fault signals is then logically OR'ed together and provided as the signal assertFault **110**.

[0022] The signal assertFault **110** is provided to an input of a latch, fault latch **112**, and to the clock input of a counter, fault counter **114**. The clear input of fault counter **114** receives a signal, P_Good, indicating whether the system power supply is operating within an acceptable voltage range. The output of fault counter **114** provides a signal latchNow **116**. The output of fault latch **112** provides a signal LATCH OUT, and is coupled to the control terminal of switch **115**. A first switch terminal of switch **115** is coupled to the nFAULT terminal **130**. A second switch terminal of switch **115** is coupled to a ground terminal.

[0023] A resistor R.sub.ext **128** is coupled between the nFAULT terminal **130** and an external voltage source V.sub.ext **126**. A first voltage source, latch supply **122**, has an input coupled to enable terminal EN **118**, and an output coupled to the supply terminal of fault latch **112**. A second voltage source **124** has an input coupled to an input voltage terminal VIN **120**, and an output coupled to the supply terminal of fault latch **112**. Voltage source **124** has an enable terminal coupled to the output of fault counter **114** that receives the signal latchNow **116**.

[0024] Safety disconnect circuit **100** automatically performs a retry following an initial fault reported by one or more of fault signals fault1 **102**, fault2 **104**, and fault3 **106** by resetting fault latch **112**. The failing safety parameter can then be retested and the fault latch **112** latches again if a safety parameter is still failing. This cycle will continue until the selected maximum number of retries is reached by the fault counter **114**, at which time the fault counter **114** will issue a latchNow signal **116** at its output, and the system will remain latched out. The downstream electronics may then be disconnected from the circuit to prevent damage to them. The maximum number of automatic retries or retests allowed can be hardwired into fault counter **114** or be customizable

according to the application. This improves the system robustness against noise and non-persistent or intermittent transient conditions.

[0025] Three safety monitors (not shown) may monitor different parameters of a DC-DC converter and generate fault signals fault1 **102**, fault2 **104**, and fault3 **106**. Fault signals fault1 **102**, fault2 **104**, and fault3 **106** are inputs to logic circuit **108**. Each fault signal is provided to a respective timing circuit within logic circuit **108** that is specific to the parameter being monitored. This helps ensure that the fault condition persists long enough before validating the fault.

[0026] If any of the fault signals results in a validated fault detection, the output of logic circuit **108** provides a signal, assertFault **110**, to the input of fault latch **112** and to the clock input of fault counter **114**, which then increments its count by one. Fault latch **112** is powered from the enable path originating from the enable terminal EN **118** for as long as the maximum count for fault counter **114** has not been reached. However, after the maximum count has been reached in fault counter **114**, fault latch **112** will be powered by an alternate supply, voltage source **124**.

[0027] Once the maximum count is reached in fault counter **114**, the signal latchNow **116** is asserted, turning on voltage source **124** and turning off latch supply **122**. In at least one case, the maximum fault count is set at four. However, the maximum fault count can instead be set to any positive integer less than or greater than four, and may be selectable based on the particular application.

[0028] Setting fault latch **112** asserts the signal LATCH OUT, which turn on switch **115**. In at least one example, switch **115** is a field effect transistor (FET) in which the gate is coupled to the output of fault latch **112**, the source is coupled to the ground terminal, and the drain is coupled to the nFAULT terminal **130**. Turning on switch **115** brings the voltage at the nFAULT terminal **130** down to ground. When switch **115** is turned off, the nFAULT terminal **130** is pulled up by the external voltage source V.sub.ext **126** through resistor R.sub.ext **128**.

[0029] To initiate a retry following a validated fault, the signal at the nFAULT terminal **130** goes low, triggering the enable of latch supply **122** through the enable terminal EN **118**. This is an indication to the system that a fault was observed in the system and, in some cases, power to the rest of the system may be shut down to avoid damage to downstream electronics. Toggling the enable terminal EN **118** clears the fault latch **112**.

[0030] Clearing fault latch **112** triggers the voltage converter to restart and commences a retest to determine whether the previous fault is still present. In many cases, power to the rest of the system remains turned off during the retests to protect the downstream electronics. If the safety fault is no longer detected, the system resumes operation. If the safety fault occurs again and a fault is observed, the fault counter **114** increases by one. This cycle of automatic retries continues until the maximum allowed faults have occurred.

[0031] Once the maximum allowed fault count is reached, the latch supply **122** is disabled and voltage source **124** is enabled by the latchNow signal **116**. When this occurs, the voltage supply for fault latch **112** switches from the enable terminal EN **118** to input voltage terminal VIN **120**, permanently latching the fault state. Toggling the enable terminal EN **118** will no longer clear the fault latch **112** when it is permanently latched. When this occurs, fault latch **112** can only be cleared by toggling the voltage at input voltage terminal VIN **120**, which also clears the fault counter **114**. Toggling the voltage at input voltage terminal VIN **120** requires removing the power supply from input voltage terminal VIN **120**, then restoring the power supply to input voltage terminal VIN **120**.

[0032] The automatic retry functionality of safety disconnect circuit **100** is enabled by toggling the dual power path to the fault latch **112** made up of latch supply **122** and voltage source **124**. In at least one implementation, the enable terminal EN **118** is controlled as a function of the signal at the nFAULT terminal **130** by external circuitry at the system level. In another implementation, the enable terminal EN **118** is controlled as a function of the signal at the nFAULT terminal **130** by an internal circuit within the voltage converter IC.

[0033] Toggling the enable terminal EN **118** clears the fault latch **112** as long as the maximum

number of retries has not yet been reached. After the fault counter **114** reaches the maximum limit of retries, the voltage supply for fault latch **112** is switched to the always-on supply rail from input voltage terminal VIN **120**, and the latched state is permanently locked in. Power remains disconnected from downstream electronics in the system during the retries, and remains disconnected if the latched state becomes permanently locked in. Toggling power to input voltage terminal VIN **120** is required in order to clear the fault latch **112** after the fault latch has been permanently locked in. The supply source for fault latch **112**, VDD_latch, is multiplexed between latch supply **122** and voltage source **124** providing a low quiescent current input. The signal latchNow **116** controls whether the supply source for fault latch **112**, VDD_latch, comes from latch supply **122** or voltage source **124**.

[0034] Safety disconnect circuit **100** controls the latching of fault latch **112** from the enable terminal EN **118**, which can be connected to a Control Area Network (CAN) bus and may receive its signal from a microcontroller via the CAN bus. This allows that function to be combined with other microcontroller controls. In some cases, the external voltage source V.sub.ext **126** is derived from an automobile battery voltage, so the voltage at the nFAULT terminal **130** can go as high as the battery voltage, which may be 15V, 24V, or 65V in some cases. The high voltage nature of the nFAULT terminal **130** allows it to operate in the same voltage domain as the battery voltage and the enable terminal EN **118**.

[0035] FIG. **2** shows a schematic diagram for an example circuit **200** for toggling the enable terminal of a safety disconnect circuit in response to the nFAULT signal. Voltage converter **202** has an enable terminal EN **118** and an nFAULT terminal **130**. Resistor **206** is coupled between the nFAULT terminal **130** and a ground terminal. Switch Q.sub.n **208** is coupled between the enable terminal EN **118** and switch Q.sub.p **212**. In at least one example, switch Q.sub.n **208** is a npn bipolar transistor. Resistor **204** is coupled between the enable terminal EN **118** and the positive terminal of a battery **220**. The negative terminal of battery **220** is coupled to the ground terminal. Battery **220** may be replaced with a different type of DC voltage source.

[0036] Resistor **210** has a first terminal coupled to the control terminal of switch Q.sub.n **208**. Switch Q.sub.p **212** is coupled between a second terminal of resistor **210** and the positive terminal of battery **220**. In at least one example, switch Q.sub.p **212** is a pnp bipolar transistor. Resistor **214** is coupled between the control terminal of switch Q.sub.p **212** and the nFAULT terminal **130**. Resistor R.sub.flt **216** is coupled between the positive terminal of battery **220** and the nFAULT terminal **130**. Capacitor C.sub.flt **218** is coupled between the nFAULT terminal **130** and the ground terminal.

[0037] If a fault is detected and validated in the system, the nFAULT terminal **130** transitions from a logic high to a logic low. The nFAULT terminal **130** transitioning low pulls the voltage at the control terminal of switch Q.sub.p **212** low and turns on switch Q.sub.p **212**. Turning on switch Q.sub.p **212** pulls the control terminal of switch Q.sub.n **208** high, and switch Q.sub.n **208** is turned on. Turning on switch Q.sub.n **208** pulls the voltage at the enable terminal EN **118** low, disabling voltage converter **202**. The timing of this is set by the resistance of resistor R.sub.flt **216** and the capacitance of capacitor C.sub.flt **218**, which determines the duration that the nFAULT terminal **130** remains low, and the duration that the enable terminal EN **118** remains low.

[0038] In response to the enable terminal EN **118** transitioning low, the voltage converter **202** clears, which causes the nFAULT terminal **130** to release and transition high. The voltage at the nFAULT terminal **130** is then pulled back up to the battery voltage with a time constant that is determined by the resistance of resistor R.sub.flt **216** and the capacitance of capacitor C.sub.flt **218**. This is merely one example implementation of a circuit for toggling the enable terminal EN **118** as a function of the nFAULT terminal **130**, and other implementations for performing the same function are also contemplated.

[0039] FIG. **3** shows a schematic diagram for an example safety disconnect circuit **300** wherein the enable terminal EN **118** toggles in response to the signal at the nFAULT terminal **130**. Fault sensors

(not shown) monitor particular parameters such as voltages, currents, and temperatures. Each fault sensor provides a respective fault signal that must be validated to ensure that the fault is not due merely to a transient condition. The validation of the faults can occur in a logic circuit (not shown) that includes logic and timing circuitry. The logic circuit (not shown) provides a signal assertFault **110** at its output that is asserted if any validated fault signal indicates a fault condition in the system.

[0040] The signal assertFault **110** is provided to an input of fault latch **112**, and to the clock input of fault counter **114**. The clear input of fault counter **114** receives a signal, P_Good, indicating whether the system power supply is operating within an acceptable voltage range. The output of fault counter **114** provides a signal latchPermanently **316**. Fault latch **112** has a power terminal coupled to the latchSupply terminal **340**. The output of fault latch **112** provides a signal LATCH OUT, and is coupled to the control terminal of switch **115**.

[0041] A supply generator **334** has an input coupled to the input voltage terminal VIN **120**, and has an output, counterSupply **336**, coupled to the supply terminal of fault counter **114**. In at least one example, supply generator is a buck voltage regulator. The voltage at the input voltage terminal may be as high as 65V, which would be harmful to provide to fault counter **114**. So, supply generator **334** takes a higher voltage from the input voltage terminal VIN **120** and brings it down to a lower voltage. A first switch terminal of switch **115** is coupled to the nFAULT terminal **130**. A second switch terminal of switch **115** is coupled to a ground terminal. A resistor R.sub.ext **128** is coupled between the nFAULT terminal **130** and an external voltage source V.sub.ext **126**.

[0042] Switch **336** is coupled between the enable terminal EN **118** and the latchSupply terminal **340**. Switch **338** is coupled between the input voltage terminal VIN **120** and the latch Supply terminal **340**. The control terminals of switch **336** and switch **338** are each coupled to the output of fault counter **114** and receive the signal latchPermanently **316**. The control terminal of switch **336** has an opposite logic from the control terminal of switch **338** so that only one of switch **336** and **338** may be closed at a time, but not both. Switches **336** and **338** operate as a multiplexer to choose the voltage source for the latchSupply terminal **340** from either the enable terminal EN **118** or the input voltage terminal VIN **120**.

[0043] Whenever a fault sensor determines that a safety limit has been violated, a fault signal is generated. If a fault signal is validated, then the signal assertFAULT **110** transitions from low to high and advances a count in the fault counter **114** by one. The signal assertFAULT **110** transitioning high also sets the fault latch **112**, causing the LATCH OUT signal to go high. Switch **115** turns on in response to the LATCH OUT signal transitioning high, causing the nFAULT terminal **130** to be connected to ground.

[0044] As long as the fault counter **114** does not reach the particular limit set for maximum number of retries following an initial failure, the signal latchPermanently **316** remains low. As long as the signal latchPermanently remains low, switch **336** is closed and switch **338** is open. So, the latchSupply terminal **340** is coupled to the enable terminal EN **118** until the signal latchPermanently **316** goes high because switches **336** and **338** are multiplexing between the enable terminal EN **118** and the input voltage terminal VIN **120**, and the enable terminal EN **118** is currently the selected input.

[0045] If the maximum fault count is reached by fault counter **114**, then the signal latchPermanently transitions from low to high, which opens switch **336** and closes switch **338**, flipping the multiplexer output. Closing switch **338** shorts the latchSupply terminal **340** to the input voltage terminal VIN **120**. The supply terminal of fault latch **112** is no longer being supplied by the enable terminal EN **118**, but is now being supplied by the input voltage terminal VIN **120**.

[0046] After this occurs, any further toggling of the enable terminal EN **118** cannot clear the fault latch **112**. In this case, the fault latch **112** can now only be cleared by removing power to input voltage terminal VIN **120**, which in some cases is at the battery voltage. In this case, the battery would have to be disconnected from the system to reset the fault latch. Disconnecting the battery

from the system requires substantially more than sending a reset signal from a microcontroller to the device through the CAN bus to reset fault latch **112** as can be done following a fault as long as the maximum fault count has not yet reached by fault counter **114**.

[0047] Prior to reaching the maximum fault count in fault counter **114**, toggling the signal at the enable terminal EN **118** clears fault latch **112** because the power supply for fault latch **112** is coming from the enable terminal EN **118**. Clearing the fault latch **112** allows the converter to restart, and allows a retest. However, this does not clear fault counter **114**, so fault counter **114** continues to increment by one each time this occurs. Removing power from the input voltage terminal VIN **120** then reapplying power resets fault latch **112** and resets fault counter **114**.

[0048] In this description, “terminal,” “node,” “interconnection,” “lead” and “pin” are used interchangeably. Unless specifically stated to the contrary, these terms generally mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device, or other electronics or semiconductor component.

[0049] In this description, “ground” includes a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground and/or any other form of ground connection applicable to, or suitable for, the teachings of this description.

[0050] In this description, the term “couple” may cover connections, communications or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal generated by device A.

[0051] In this description, even if operations are described in a particular order, some operations may be optional, and the operations are not necessarily required to be performed in that particular order to achieve specified results. In some examples, multitasking and parallel processing may be advantageous. Moreover, a separation of various system components in the embodiments described above does not necessarily require such separation in all embodiments.

[0052] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A safety disconnect circuit, comprising: a logic circuit having first, second and third logic inputs and a logic output; a latch having a latch input, a latch power input, and a latch output, wherein the latch input is coupled to the logic output; a counter having a counter input and a counter output, wherein the counter input is coupled to the logic output; a switch coupled between a fault terminal and a ground terminal and having a switch control terminal, wherein the switch control terminal is coupled to the latch output; a first voltage supply having a first supply input and a first supply output; a second voltage supply having a second supply input and a second supply output; and a multiplexer having first and second multiplexer inputs, a multiplexer control terminal, and a multiplexer output, wherein the first multiplexer input is coupled to the first supply output, the second multiplexer input is coupled to the second supply output, the multiplexer output is coupled to the latch power input, and the multiplexer control terminal is coupled to the counter output.
2. The safety disconnect circuit of claim 1, wherein the first supply output provides a lower voltage to the latch power input if the first supply output is selected by the multiplexer, and the second supply output provides a higher voltage to the latch power input if the second supply output is selected by the multiplexer.
3. The safety disconnect circuit of claim 2, wherein the multiplexer selects the first supply output if a count of the counter has not reached a maximum count value, and selects the second supply

output if the count has reached the maximum count value.

4. The safety disconnect circuit of claim 3, wherein toggling the first supply output resets the latch if the count has not reached a maximum count value, and does not reset the latch if the count has reached the maximum count value.

5. The safety disconnect circuit of claim 4, wherein toggling the second supply output resets the latch and resets the count to zero.

6. The safety disconnect circuit of claim 1, further comprising an external voltage supply coupled to the fault terminal through a resistor.

7. The safety disconnect circuit of claim 3, wherein the second supply input is coupled to an automobile battery.

8. The safety disconnect circuit of claim 1, wherein each of the first, second and third logic inputs are provided by a respective fault detection circuit.

9. The safety disconnect circuit of claim 8, wherein the logic circuit includes timing circuits, and each of the first, second and third logic inputs are coupled to a respective timing circuit.

10. The safety disconnect circuit of claim 1, wherein the first voltage supply is controlled by a signal provided at the fault terminal.

11. A safety disconnect circuit, comprising: a latch having a latch input, a latch power input, and a latch output, wherein the latch input is coupled to a validation terminal; a counter having a counter input, a counter supply terminal and a counter output, wherein the counter input is coupled to the validation terminal; a first switch coupled between a fault terminal and a ground terminal and having a first switch control terminal, wherein the first switch control terminal is coupled to the latch output; a second switch coupled between a first voltage supply terminal and the latch power input, and having a second switch control terminal, wherein the second switch control terminal is coupled to the counter output; and a third switch coupled between a second voltage supply terminal and the latch power input, and having a third switch control terminal, wherein the third switch control terminal is coupled to the counter output.

12. The safety disconnect circuit of claim 11, wherein the first voltage supply terminal provides a lower voltage than the second voltage supply terminal, and the second voltage supply terminal is coupled to an automobile battery.

13. The safety disconnect circuit of claim 11, wherein the second switch is closed and the third switch is open responsive to a count of the counter not having reached a maximum count value, and the second switch is open and the third switch is closed responsive to the count having reached the maximum count value.

14. The safety disconnect circuit of claim 13, wherein toggling power provided to the first voltage supply terminal resets the latch if the count has not reached a maximum count value, and does not reset the latch if the count has reached the maximum count value.

15. The safety disconnect circuit of claim 14, wherein toggling power provided to the second voltage supply terminal resets the latch and resets the count to zero.

16. The safety disconnect circuit of claim 11, further comprising an external voltage supply coupled to the fault terminal through a resistor.

17. The safety disconnect circuit of claim 11, wherein the second voltage supply terminal is coupled to an automobile battery.

18. The safety disconnect circuit of claim 11, wherein a signal provided at the validation terminal indicates a system fault has been detected.

19. The safety disconnect circuit of claim 11, further comprising a voltage regulator having a regulator input and a regulator output, wherein the regulator input is coupled to the second voltage supply terminal, and the regulator output is coupled to the counter supply terminal.

20. The safety disconnect circuit of claim 11, wherein a voltage at the first voltage supply terminal is controlled by a signal provided at the fault terminal.
